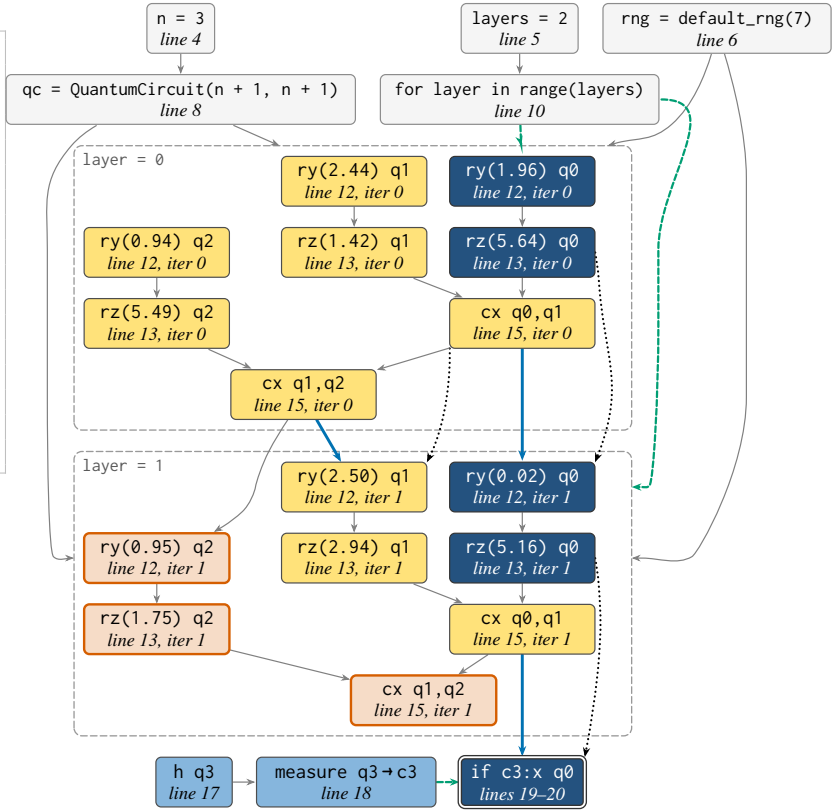
(a) VQE ansatz

```
4  n = 3
5  layers = 2
6  rng = np.random.default_rng(7)
7
8  qc = QuantumCircuit(n + 1, n + 1)
9
10 for layer in range(layers):
11     for qubit in range(n):
12         qc.ry(float(rng.uniform(0, np.pi)), qubit)
13         qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14     for qubit in range(n - 1):
15         qc.cx(qubit, qubit + 1)
16
17 qc.h(n)
18 qc.measure(n, n)
19 with qc.if_test((qc.clbits[n], 1)):
20     qc.x(0)
```

- data, classical or quantum
- classical control, loop or measured bit
- quantum via a control, not in classical PDG
- classical PDG and def-use only, skips a CNOT
- kept by every rule, def-use included
- added by the classical PDG
- added by the QDG
- added by the QPDG, not a dependence
- last write to the criterion
- classical statement, all in the QDG slice
- one iteration of the outer loop, with the inner loops of lines 11 and 14 unrolled

(b) Dependence graph of the four rules, criterion q_0



slices of q_0, of 19 operations:
def-use 5, $F = 0.55$ · classical PDG 7, $F = 0.97$
QDG 16, exactly the dependences, $F = 1.00$ · QPDG 19, $F = 1.00$